Mechanical properties of Pb/Sn Pb/In and Sn-In solders

W. Kinzy Jones, Yanqing Liu, Milind Shah
Department of Mechanical Engineering, Florida International University, Florida, USA
Robert Clarke
Teledyne Electronic Technologies, Los Angeles, California, USA

The mechanical properties of eight solder alloys from the Pb-Sn-In-Ag alloy systems were determined over the temperature range –200°C to 100°C, using uniaxial tensile tests, dynamic mechanical analysis (DMA), acoustic pulse methods and dilatometry. In general, the strength and elastic modulus of the alloys studied was inversely dependent on temperature. PbSn, PbIn and SnIn alloys were observed to turn superplastic with elongations over 100 per cent at temperatures of 50°C or above. The Pb-based and In-Sn eutectic solders possessed superplasticity at temperatures greater than 50°C. From these results, deformation and fracture processes are reviewed, and the appropriate fracture mechanism is proposed.

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Introduction

The introduction of newer materials for thermal management, superconductors, large area dies, flip-chip assembly, and multi-layer, multi-material structures in MCM fabrication, have placed increased demands for usage and characterization of low temperature joining alloys[1-4]. The thermal cycling in these processes can lead to thermal fatigue, and eventually, to solder joint failure. Considerable effort has been made to study thermal-fatigue-induced shear and to evaluate solder joint reliability in those solder alloys most popular in electronic packaging[5-7]. The reliability of any solder joint interconnection depends on the resilience of the materials and joint structure on the stress and strain created by thermal mismatch. Ultimately, it is thermal fatigue resistance which dictates the ideal geometry and choice of material for the interconnection structure. Although there have been steady advances in computational modeling tools which can accommodate nonlinear material responses, accurate modeling is only possible when material properties are available over the entire desired temperature range. In most cases, the material property data are limited. Hence, the room temperature values of the required properties are used (e.g. elastic modulus, thermal expansion, plasticity and strength, Poisson ratio) resulting in model predictions which are often of little value other than to indicate trends. In order to understand the change in mechanical properties, it is necessary to study the deformation and fracture behavior of solders at the temperatures of interest[8,9].

In this study, the mechanical properties of eight Pb-Sn, In-Sn, In-Pb solders were evaluated over the temperature range from –200°C to 100°C. The deformation and fracture process was investigated, and their fracture mechanism is proposed. According to the matrix type of the alloys, the solder alloys can be broken into two groups:

1. eutectics alloys (63Sn/37Pb, 62Sn/36Pb/2Ag, 96Sn/4Ag, 49In/48.5Sn/2.5Ag, and 50In/50Sn);
2. solid solutions alloys (50In/50Pb, 5Sn/95Pb and 10Sn/90Pb).

Experimental procedures

Commercially available eutectic solders (95Pb/5Sn, 90Pb/10Sn, 63Sn/37Pb, 62Sn/ 36Pb/2Ag, 96Sn/4Ag, 50In/50Pb and 50In/50Sn) were used in the experiment. One alloy solder (49In/48.5Sn/2.5Ag) was melted and cast into ingot by mixing the In-Sn eutectic solder with pure Ag.

The ingot solder was melted and cast into a water-cooled aluminum mold. The mass of the mold, small sample size and water-cooling ensured a fine-grained sample with a grain structure similar to that found in solder joints.

Tensile tests were carried out on an Instron Model 1011 tensile testing machine. The strain rate of 0.003 sec^{-1} was used for this work. The tensile samples had a total length of 40mm, a gage length of 15mm and a thickness of 2mm. Three samples per test configuration were used. In order to perform testing over the range from –200°C to 100°C, the tensile tester was fitted with a thermally insulated container and special, small grips. The sample was immersed in a solution of liquid nitrogen/alcohol for low temperatures or heated glycerin for high temperatures. Sample temperatures were monitored by a thermocouple.

Following tensile testing, sample surfaces and sample fractures were characterized using scanning electron microscopy (SEM) for fracture mode. SEM observations were made directly on the fracture surfaces and also on cross-sectioned samples in which the sectioned surface was taken normal to the fracture surface. Energy dispersive X-ray spectroscopy (EDS) was used to determine the composition and to distinguish among different phases on the fracture surface.

Test results

Typical eutectic structure was observed in the 50In/50Sn solder, as shown in Figure 1(a). It appears that the Sn-rich phase was deposited in the In-rich phase matrix. The size of the eutectic colony was in the range of 100-200μm. EDS analysis indicated that the In-rich phase contained 29 percent Sn, while the Sn-rich phase contained 27 percent In.

Figure 1(b) shows the microstructure of as-melted alloying solders which present different amounts and morphology of the second phase. Fine dendritic Ag_2In and accicular $AgIn_2$ were observed in the solder. These two phases are intermetallic solid solution with variable atom ratio; some of the Sn atoms are dissolved in this phase.

Metallurgical graphic analysis shows that the 50In/50Pb solder is single-phase solid solution. There are secondary phase particles (Sn) in the Pb-rich matrix for 95Pb/5Sn and 90Pb/10Sn.

The results of the uniaxial tension tests are shown in Figures 2 to 6. The following observations were made:

1. For 50In/50Pb, 95Pb/5Sn and 90Pb/10Sn, the strength and the elongation change linearly with the temperature.
2. For 96Sn/4Ag, the strength increases linearly with decreasing temperature above –50°C, but increases sharply below –100°C. A peak appears at about –125°C. The elongation decreases with decreasing temperature above –50°C; below –100°C it drops sharply.
3. For 63Sn/37Pb, the strength and elongation are similar to 96Sn/4Ag at below room temperature, however, superplasticity is exhibited above 50°C.
4. In-Sn eutectic solders possess higher levels of uniform elongation at low temperatures and alloying can improve strength still further.

The elastic modulus for the five Pb-Sn solders using the acoustic method are shown in Figure 7. The elastic modulus increases with decreasing temperature.

Discussion

Plasticity versus temperature of Pb-Sn solders

The elongation is a main index of material plasticity. The test results of the elongation, including total elongation δ_T, and the uniform elongation δ_U are shown in Figures 3 and 6, respectively.

The total elongation δ_T increases with the temperature for the solders as shown in Figure 3. For 95Pb/5Sn and 90Pb/10Sn, the δ_T slowly increases linearly with temperature, where the other solders do not exhibit a linear relationship. Typically, three steps appear on the δ_T curve of 63Sn/37Pb:

1. above 50°C, the δ_T rises rapidly;
2. below –100°C, it drops sharply; and
3. between –100°C and 50°C, it increases linearly and slowly.

W. Kinzy Jones, Yanqing Liu, Milind Shah and Robert Clarke
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Soldering & Surface Mount Technology
10/1 [1998] 37–41

Figure 1
The microstructure of (a) 50In/50Sn and (b) second phase in as-melted solder

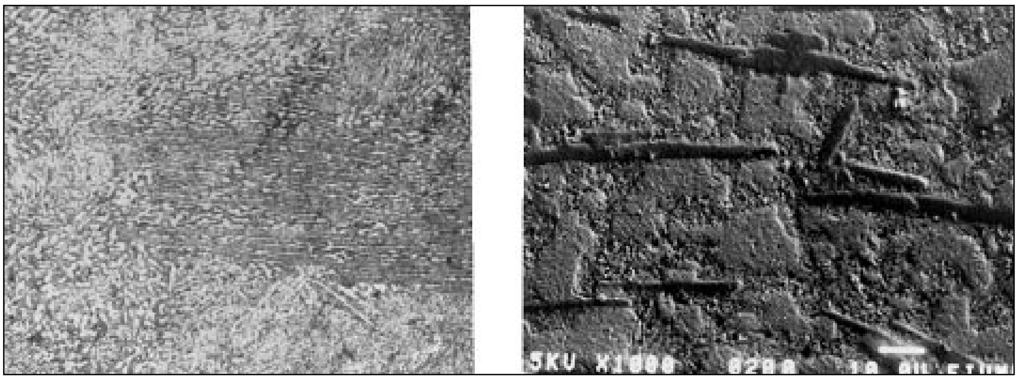


Figure 2
UTS versus temperature in Pb-Sn solders

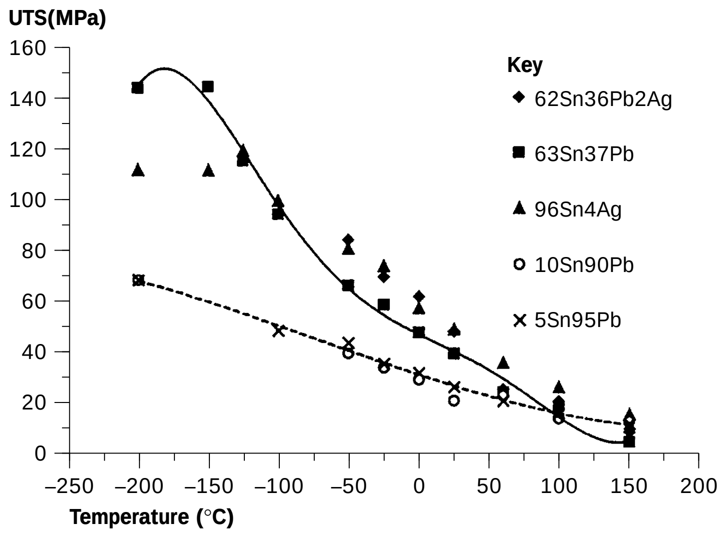
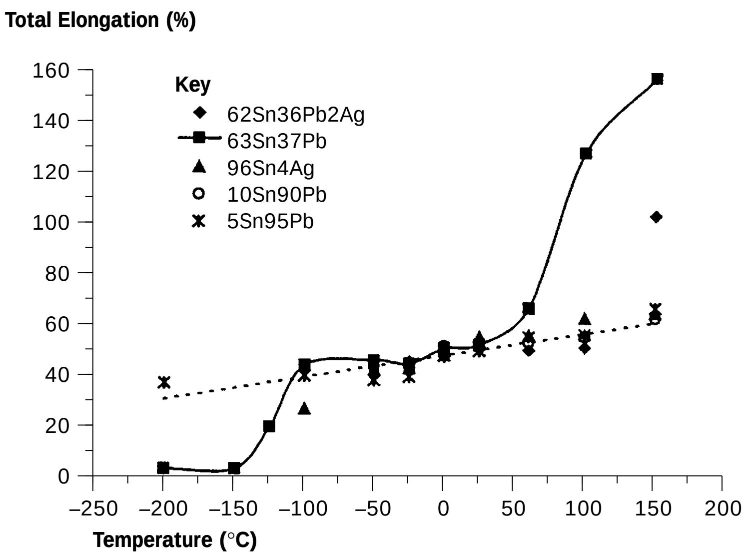


Figure 3
Total elongation versus temperature in Pb-Sn solders



W. Kinzy Jones, Yanqing Liu, Milind Shah and Robert Clarke
Mechanical properties of Pb/Sn Pb/In and Sn-In solders

Soldering & Surface Mount Technology
10/1 [1998] 37–41

Figure 4
UTS versus temperature in In-Pn and In-Sn solders

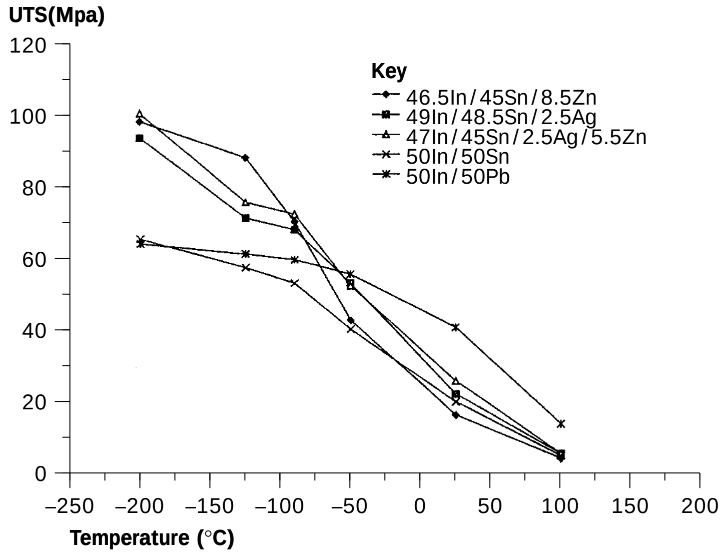


Figure 5
Total elongation versus temperature in In-Pn and In-Sn solders

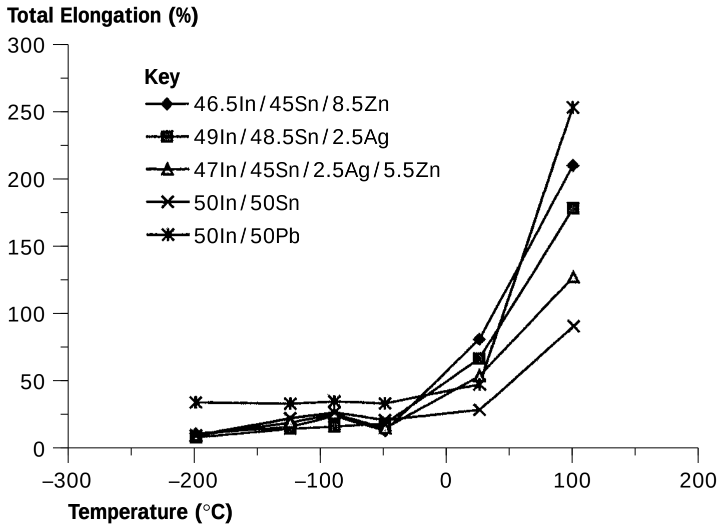
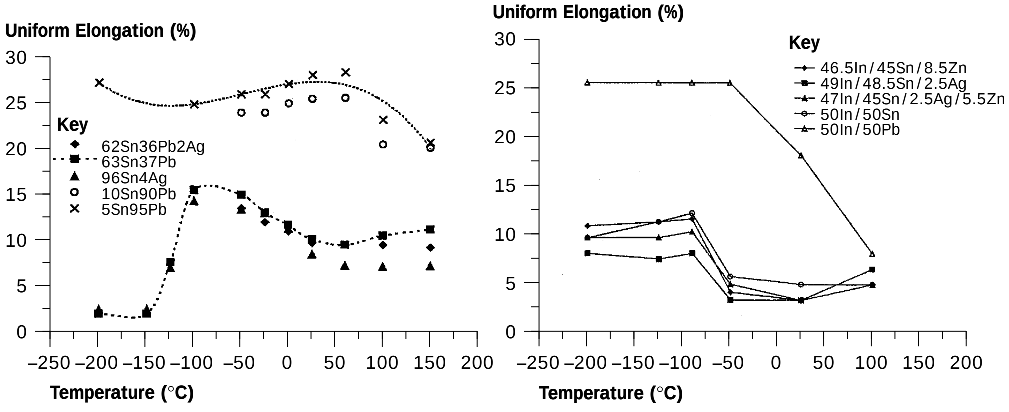


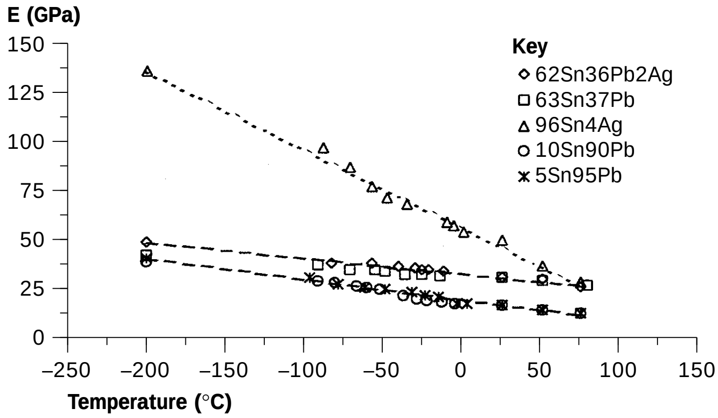
Figure 6
Uniform elongation versus temperature in the test solders



W. Kinzy Jones, Yanqing Liu,
Milind Shah and Robert Clarke
Mechanical properties of Pb/Sn Pb/In and Sn-In solders

Soldering & Surface Mount Technology
10/1 [1998] 37–41

Figure 7
Elastic modulus versus temperature in Pb-Sn solder



The causes of the change in the δ_T are the different mechanisms for deformation and fracture. Fracture analysis indicates that the three stages correspond to the following fracture mechanisms.

Below –150°C

Brittle fracture occurs below this temperature with the highest strength and lowest elongation in the tensile test. All cleavage facets consist of the dimples in the test samples. This "quasi-cleavage" feature appears on the fracture surface, as shown in Figure 8. The forming mechanism of the dimples in the solder suggests that at this temperature the microcracking was initiated at the cleavage of the Sn-rich phase due to low plasticity, and then the tearing ridge formed as plastic deformation in the surrounding Pb-rich phase, as illustrated in Figure 9.

Between –100°C and 50°C

Ductile fracture occurred at this temperature range. The strength and elongation of the alloys changed linearly with decreasing temperature. Plastic deformation was mainly accommodated by slip. The dimple features appeared on the fracture surface in the tested solders.

Above 50°C

Superplastic deformation occurred in the 63Sn/37Pb and 62Sn/36Pb/2Ag samples at this temperature range. At 150°C, when the temperature is close to the melting point of the alloys, the plastic deformation is accommodated by the sliding of phase boundaries in the eutectic colonies. The fracture features show that the fracture is caused by sliding separation along the interface between both phase particles (Figure 10).

Figure 8
"Quasi-cleavage" fracture at –200°C in 63Sn/37Pb

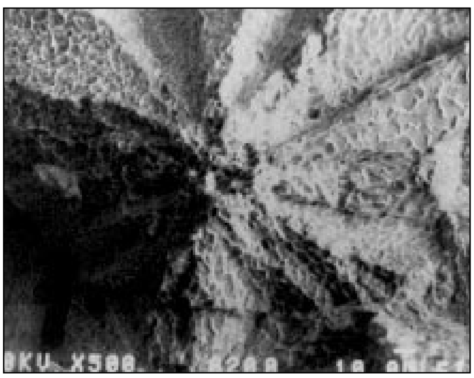


Figure 9
Schematic illustrating the formation of "quasi-cleavage" fracture

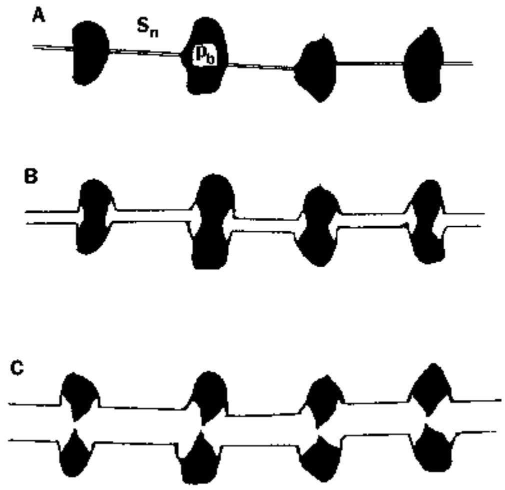
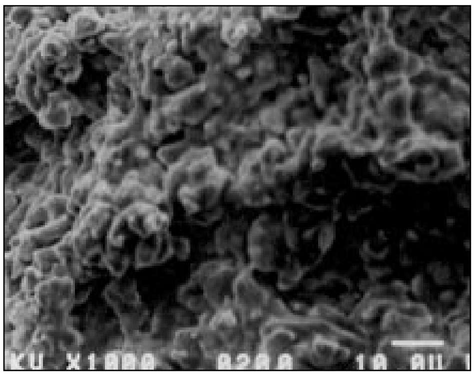


Figure 10
Fracture surface of 63Sn/37Pb at 100°C



It is interesting that the 96Sn/4Ag solder displays a duality between the solid solution alloys (such as 95Pb/5Sn) above 50°C (no superplasticity) and the typical eutectic alloys (such as 63Sn/37Pb) below –100°C (brittle fracture). Brittle fracture is sensitive to the crystal structure of the alloys, and the superplasticity is sensitive to the microstructure. At high

W. Kinzy Jones, Yanqing Liu, Milind Shah and Robert Clarke
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Soldering & Surface Mount Technology
10/1 [1998] 37–41

temperatures, the 96Sn/4Ag has a crystal structure that approaches a solid solution, with finely dispersed Ag particles, due to the low concentration of silver particles predicted by the lever law. This structure inhibits superelastic response. However, at low temperatures, the tetragonal structure of Sn promotes brittle fracture by cleavage.

Uniform elongation

As shown in Figure 6, the δ_U possesses the following features:

- the Pb solid solution solders (95Pb/5Sn, 90PB/10Sn) possess higher δ_U until –200°C; however, Sn matrix eutectic solders have lower δ_U and drop sharply below –100°C;
- a peak appears at approximately $T/Tm = 0.5$ for Pb-Sn eutectic solders.

For In-Sn eutectic solders, the uniform elongation doubles at the temperature close to 0.5*Tm*, and then remains constant until –200°C.

The uniform elongation indicates a capability for both uniform plastic deformation and strengthening in the matrix phase. The FCC Pb-matrix alloys possess a higher capability of strengthening. Meanwhile, the different fracture mechanisms for the two types of solder at low temperatures results in the difference in the δ_U, as previously described. The peak of the δ_U results from the interaction between work-hardening and recovering at this temperature range.

Low temperature plasticity of In-Sn solders

In-Sn eutectic solders underwent a moderate amount of plastic deformation, even at temperatures under –200°C, as shown in Figure 6. However, Pb-Sn eutectic solders fractured without any evident plasticity at the same temperature[10]. This was caused by different fracture mechanisms at low temperature. The equilibrium phase diagrams of In-Sn and Pb-Sn binary system indicate that the eutectic structure of the In-Sn system consists of two intermetallic phases, β (In-rich phase) and γ (Sn-rich phase), while that of the Pb-Sn system consists of two solid solutions. Due to the immiscibility of Pb and Sn, the two solid solutions dissolved very limited amounts of external, foreign atoms at room temperature. To some extent, they were supposed to behave as pure Pb and Sn at low temperature (<–50°C). Sn was prone to cleavage and thus resulted in brittle fracture of the Pb-Sn solder, even though Pb deformed plastically[11]. The structures of the β- and γ- phases in In-Sn eutectic solders were quite distinct from those of In and Sn. No cleavage features were detected in the fracture surfaces, indicating that both phases of β and γ deformed plastically even under very low temperature. Further research work is necessary to reveal the micromechanism of the plasticity in the γ-phase.

Conclusions

1. For Pb-rich and In-Pb solid solution solders (95Pb/5Sn, 90Pb/10Sn and 50In/50Pb), the linear relationship of the mechanical properties versus the temperature is caused by the ductile fracture mechanism which does not change over the range from –200°C to 100°C.
2. For Sn-rich matrix eutectic solders, three distinct stages of elongation with temperature correspond to three different fracture mechanisms: superplastic fracture, ductile fracture and quasi-cleavage fracture.
3. The 96Sn/4Ag solder displays mechanical properties found both in 95Pb/5Sn and in 63Sn/37Pb, with the non-cubic crystal structure in the Sn matrix (same as 63Sn/37Pb) being demonstrated at low temperatures and the microstructure (same as 95Pb/5Sn) controlling the higher temperature region.
4. The 50In/50Sn eutectic solder possessed a higher level of uniform elongation at low temperatures. The strength of the In-Sn eutectic solder was improved by alloying with Ag, and also allowed higher uniform elongation at low temperature.
5. For Pb-Sn solders, the elastic modulus decreased linearly with increasing temperature until 70°C, and then rapidly dropped as temperature rose above 100°C. The elastic modulus obtained using the uniaxial tension test appeared to be highly erroneous for the solders, whereas the results of the acoustic pulse method and the DMA were in agreement.

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